

STAT 486 Project 3

Logistic Regression Analysis of 1986 Challenger

Disaster

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Abstract

In this report, observational data on O-rings, a critical shuttle seal, was examined in order to understand the effect of weather conditions on factors such as thermal distress and damage sustained to space shuttle O-rings. The data was fitted to two logistic regression models, both used to study the relationship between the two variable pairs of interest. Temperature at the time of a devastating shuttle launch in 1986 was analyzed and found to have been linked with a high probability of thermal distress, as well as a high probability of O-ring failure. The Wald test and the likelihood ratio test were both utilized for verifying the significance of the temperature variable to thermal distress, but the two tests gave differing results. The Wald test resulted in the conclusion of no significance, while the likelihood ratio test resulted in temperature being a highly significant variable. Lastly, the change in the estimated probability per degree increase in temperature was calculated for thermal distress and O-ring failure, totaling to probability values of 0.6048347 and 0.7927875 respectively.

Introduction

An investigation into the cause of the 1986 Challenger space shuttle disaster sought to examine whether the damage to critical seals called O-rings was related to the temperature at the time of shuttle launch. The aim of this report is to utilize relevant data from 23 shuttle launches in order to examine the effect of temperature on thermal distress, and subsequently, the effect of temperature on O-ring failure. Variables contained within the dataset are displayed in table 1.

Table 1: Variables

Ft	Flight number
Temperature	Temperature at time of flight
TD	Thermal distress on Oring (1 = yes, 0 = no)
Damaged	Number of O-ring damaged after shuttle launch
Undamaged	Number of O-rings undamaged after shuttle launch

Logistic regression is utilized in order to examine the relationship between notable variables. Namely, the relationship between thermal distress and temperature, as well as O-ring failure and temperature. The variable containing the data for damages is used to achieve this. The logistic regression models constructed are as follows:

Formula 1: Thermal Distress by Temperature

$$Y = \frac{e^{(32.3381 - 0.5028x_{\text{temperature}})}}{1 + e^{(32.3381 - 0.5028x_{\text{temperature}})}}$$

Formula 2: O-ring failure by Temperature

$$Y = \frac{e^{(15.0429 - 0.2322x_{\text{temperature}})}}{1 + e^{(15.0429 - 0.2322x_{\text{temperature}})}}$$

Procedure

The dataset was uploaded into Rstudio, where it was examined. A matrix of scatterplots was obtained, and the summary statistics of variable Temperature were calculated.

Figure 1: Scatterplot Matrix of dataset Orings

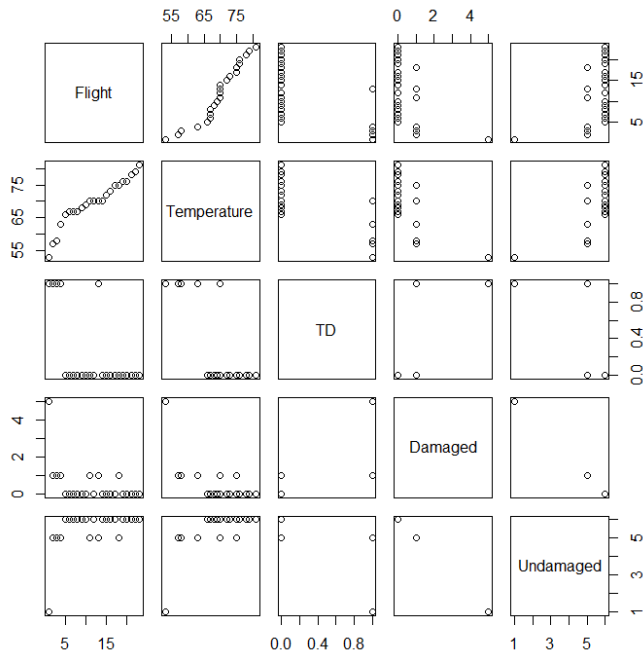
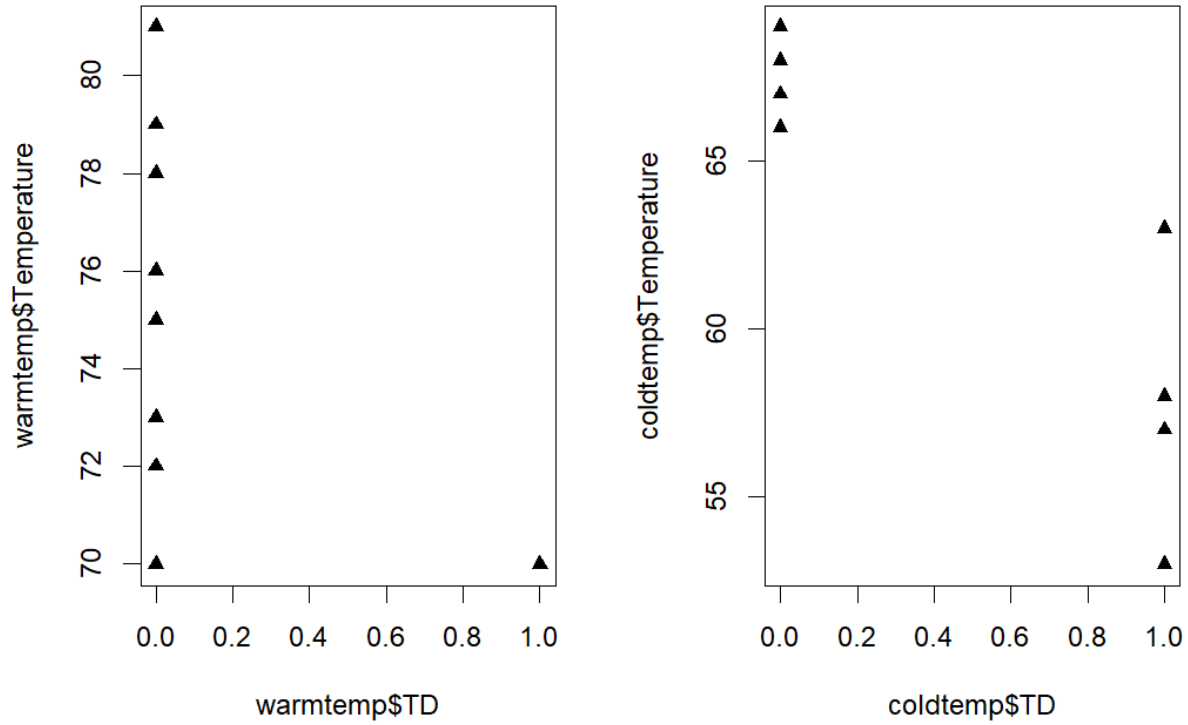


Table 2: Temperature summary statistics

Min.	1st Qu.	Median	Mean	3rd Qu.	Max
53.00	67.00	70.00	69.57	75.00	81.00

The highest recorded temperature was 81 degrees fahrenheit, while the lowest was 53 degrees. To further examine the effect of temperature on thermal distress, temperature was divided into two groups of “warm” ($\geq 70F$) and “cold” ($<70F$). Scatter plots were created to observe the spread of the data for each partition, displayed in figure 2.

Figure 2: Scatterplots of warm temp and cold temp vs thermal distress



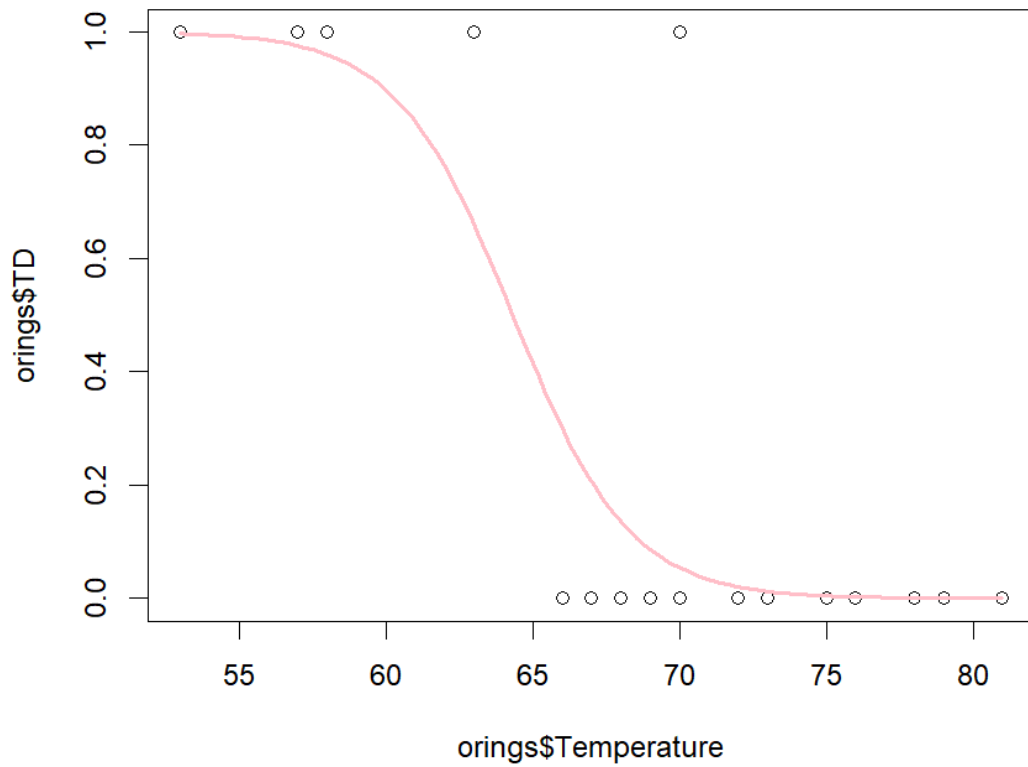
With the exception of one data point at 70F, all noted points of no thermal distress lie above the temperature of 65F. At temperatures below that value, thermal distress was recorded true.

Logistic regression was utilized in order to model the effect of temperature on the probability of thermal distress. The model created is as follows:

$$Y = \frac{e^{(32.3381 - 0.5028x_{temperature})}}{1 + e^{(32.3381 - 0.5028x_{temperature})}}$$

Figure 3 additionally displays a plot of the fitted model.

Figure 3: Logistic regression model of thermal distress by temperature



A summary of the model provides certain values of note, such as the AIC value of 13.803, and the null and residual deviance, being 24.0850 and 9.8032 respectively.

The model was used to estimate the probability of thermal distress at the time of the Challenger flight, when the temperature was 31F. The value received from this operation was a probability of 0.9999999. Based on the data collected, it was almost certain that thermal distress was suffered at the time of the flight.

In order to determine the temperature when the probability of thermal distress is at 0.5, the model equation was used.

$$\frac{\beta_0}{\beta_1} = \frac{32.3381}{-0.5028} = -64.31603$$

This value was then verified using the model equation.

$$\frac{\exp(32.3381+(-0.5028*64.31603))}{1+\exp(32.3381+(-0.5028*64.31603))} = 0.5$$

The temperature at which the probability of thermal distress is 0.5 is found to be 64.32F.

The odds ratio is calculated in order to assess the change in the estimated probability per degree increase in temperature.

$$\exp(-0.5028) = 0.6048347$$

For every one degree Fahrenheit increase in temperature, the odds of thermal distress occurring goes down by a factor of 0.6048347.

Two tests are performed to test the hypothesis that temperature has no effect on the response variable. In both cases, the null and alternative hypotheses are as follows, with a significance level of 0.05 chosen.

$$H_0: \beta_1 = 0$$

$$H_1: \beta_1 \neq 0$$

First, the Wald test is performed. This test had an F-statistic of 3.6193, and the p-value was found to be 0.07092. Due to the chosen significance level, the null hypothesis is not rejected, from which the conclusion is that there is not enough evidence to claim that temperature is very significant.

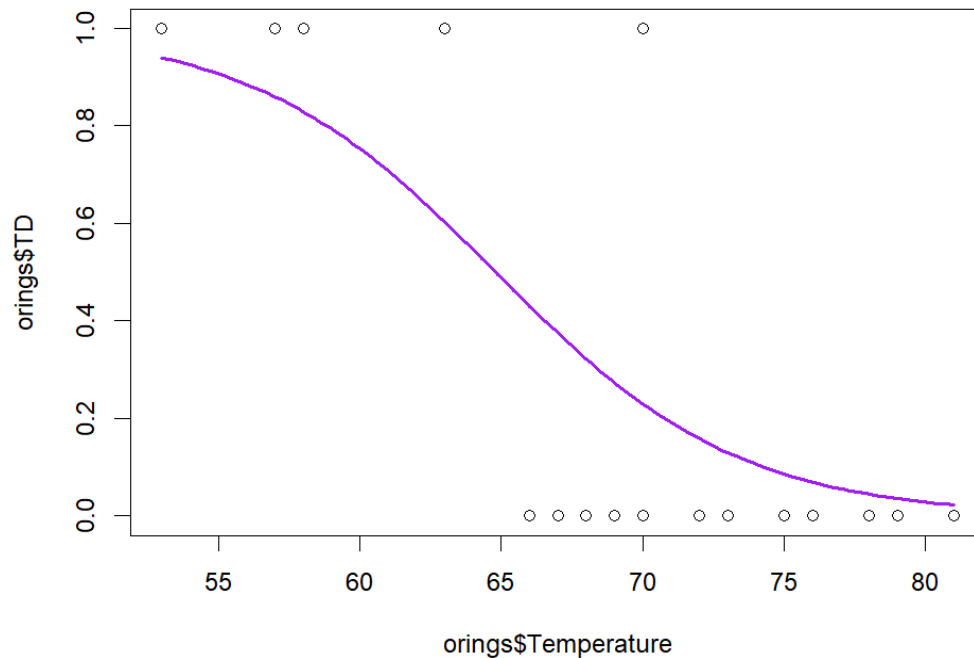
Next, the likelihood ratio test is performed. The Chi-squared statistic is 14.282, and the p-value is found to be 0.0001574. For this test, the null hypothesis is rejected, and temperature is found to be very significant.

Following analysis of thermal distress and temperature, the next focus is O-ring failures and temperature directly. Logistic regression was then utilized again in order to model the effect of temperature on the probability of O-ring failure. The model created is as follows:

$$Y = \frac{e^{(15.0429 - 0.2322x_{\text{temperature}})}}{1 + e^{(15.0429 - 0.2322x_{\text{temperature}})}}$$

The fitted model is displayed on the graph in figure 4.

Figure 4: Logistic regression model of O-ring failure by temperature



Utilizing the model to predict the probability of O-ring failure at the temperature of 31F returns a probability of 0.9996088, or a near surety of O-ring failure. At the time of the Challenger's launch, it was certain that the temperature would lead to O-ring failure.

Using the previous methods to attain the temperature associated with an O-ring failure probability of 0.5 returns a value of 64.78424, or 64.78F.

$$\frac{\beta_0}{\beta_1} = \frac{15.0429}{-0.2322} = -64.78424$$

$$\frac{\exp(15.0429 + (-0.2322 * 64.78424))}{1 + \exp(15.0429 + (-0.2322 * 64.78424))} = 0.4999999$$

The temperature at which the probability of O-ring failure is 0.50 is found to be 64.78F. Following this, the odds ratio is calculated in order to determine the change in the estimated probability per degree increase in temperature.

$$\exp(-0.2322) = 0.7927875$$

For every one degree Fahrenheit increase in temperature, the odds of O-ring failure occurring goes down by a factor of 0.7927875.

Conclusion

After concluding the analysis of the different aspects of this dataset, temperature was found to have a significant effect on both thermal distress and possibility of O-ring failure. Higher temperatures were associated with a lower chance of thermal distress, while lower temperatures seemed to indicate a higher probability of its occurrence. The temperatures of the day of the Challenger flight crash showed a near certain chance of O-ring failure, which can explain the disastrous consequences of that launch.

Appendix (R code)

```
# HEBA SYED
```

```
setwd("C:/Users/heb/Documents/STAT")
```

```
library(aod)
```

```
library(lmtest)
```

```
orings<-read.csv("Orings.csv")
```

```
summary(orings$Temperature)
```

```
# split
```

```
spl<-split(orings, orings$Temperature >= 70)
```

```
warmtemp<-spl[[2]]
```

```
coldtemp<-spl[[1]]
```

```
# plots
```

```
par(mfrow=c(1,2))
```

```
plot(warmtemp$TD,warmtemp$Temperature, pch = 17)
```

```
plot(coldtemp$TD,coldtemp$Temperature, pch = 17)
```

```
par(mfrow=c(1,1))
```

```
pairs(orings)
```

```
corrmatrix<-data.frame(round(cor(orings), 3))
```

```
plot(orings$TD ~ orings$Temperature)
```

```
plot(jitter(orings$TD, 0.1) ~ orings$Temperature)
```

```
model1 <- glm(TD ~ Temperature, data = orings, family = binomial)
```

```
summary(model1)
```

```

# Call:
# glm(formula = TD ~ Temperature, family = binomial, data = orings)
#
# Coefficients:
#   Estimate Std. Error z value Pr(>|z|)
# (Intercept)  32.3381    17.6301   1.834   0.0666 .
# Temperature  -0.5028     0.2643  -1.902   0.0571 .
# ---
#   Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#
# (Dispersion parameter for binomial family taken to be 1)
#
# Null deviance: 24.0850  on 22  degrees of freedom
# Residual deviance:  9.8032  on 21  degrees of freedom
# AIC: 13.803
#
# Number of Fisher Scoring iterations: 7

plot(residuals(model1) ~ fitted(model1),
      xlab = expression(hat(y)[i]), ylab = expression(r[i]), cex=0.8)
abline(0, 0, lty = 2, col=4)

x <- seq(min(orings$Temperature), max(orings$Temperature), length =
100)
y <- predict(model1, newdata = data.frame(Temperature = x), type =
"response")
plot(orings$Temperature, orings$TD, col = "black")
lines(x, y, col = "pink", lwd = 2)

par(mfrow=c(2,2))

```

```

plot(model1)
par(mfrow=c(1,1))

predict(model1, newdata = data.frame(Temperature = 31), type =
"response")
# 0.9999999

32.3381/-0.5028 #
# -64.31603
(exp(32.3381+(-0.5028*64.31603)))/(1+exp(32.3381+(-0.5028*64.31603)))
# 0.5

exp(-0.5028) # odds ratio
# 0.6048347

wald.test(Sigma = vcov(model1), b = coef(model1), Terms = 1)
# Wald test
#
# Model 1: TD ~ Temperature
# Model 2: TD ~ 1
# Res.Df Df      F Pr(>F)
# 1      21
# 2      22 -1 3.6193 0.07092 .
# ---
#   Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

null<-glm(TD ~ 1, data = orings, family = binomial)
lrtest(null, model1)
# Likelihood ratio test
#

```

```

# Model 1: TD ~ 1
# Model 2: TD ~ Temperature
# #Df    LogLik Df  Chisq Pr(>Chisq)
# 1     1 -12.0425
# 2     2  -4.9016  1 14.282  0.0001574 ***
#    ---
#    Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

model2<-glm(Damaged>= 1 ~ Temperature, family = binomial, data =
orings)
summary(model2)
# Call:
#   glm(formula = Damaged >= 1 ~ Temperature, family = binomial,
#       data = orings)
#
# Coefficients:
#   Estimate Std. Error z value Pr(>|z|)
# (Intercept)  15.0429      7.3786   2.039   0.0415 *
# Temperature  -0.2322      0.1082  -2.145   0.0320 *
#    ---
#    Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#
# (Dispersion parameter for binomial family taken to be 1)
#
# Null deviance: 28.267  on 22  degrees of freedom
# Residual deviance: 20.315  on 21  degrees of freedom
# AIC: 24.315
#
# Number of Fisher Scoring iterations: 5

```

```
x <- seq(min(orings$Temperature), max(orings$Temperature), length =  
100)
```

```
y <- predict(model2, newdata = data.frame(Temperature = x), type =  
"response")
```

```
plot(orings$Temperature, orings$TD, col = "black")
```

```
lines(x, y, col = "purple", lwd = 2)
```

```
predict(model2, newdata = data.frame(Temperature = 31), type =  
"response")
```

```
# 0.9996088
```

```
15.0429 / -0.2322
```

```
# -64.78424
```

```
(exp(15.0429+(-0.2322*64.78424)))/(1+exp(15.0429+(-0.2322*64.78424)))
```

```
# 0.4999999
```

```
exp(-0.2322) # odds ratio
```

```
#0.7927875
```